Dr. Gopi Raju Matta | Researcher | Computational Imaging & 3D Vision

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Research Summary

My research lies at the intersection of Computational Imaging, Computer Vision, and Machine Learning. I develop resilient radiance field representations capable of robust 3D scene reconstruction and novel view synthesis under severe optical and temporal degradations including lens flares, motion blur, sparse observations, and event-based sensing.

Research interests: Neural Scene Representations, 3D Gaussian Splatting, Neural Radiance Fields, Novel View Synthesis, Computational Photography, Event-based Vision, and Physics-based Imaging.

Education

Program	*Institution*	*CGPA/%*	*Year*
Ph.D. (Electrical Engineering)	IIT Madras	**9.0/10**	2017–2026
Thesis: *Resilient Radiance Fields: Robust 3D Scene Reconstruction Under Severe Optical and Temporal Degradations*			
Advisor: Prof. Kaushik Mitra · **Degree Awarded:** July 17, 2026			
M.Tech. (Electronics & Telecommunication Engg.)	IIEST Shibpur	**84.4%**	2015–2017
B.Tech. (Electronics & Communication Engg.)	RGUKT, IIIT Nuzvid	**9.3/10**	2011–2015

Publications

- 1) **G. R. Matta**, T. Reddypalli, D. M. Vemunuri, and K. Mitra, "PRISM3D: Probabilistic Refinement and Robust Initialization for Physically Consistent Scene Modeling under Extreme Motion Blur," *Accepted to European Conference on Computer Vision (ECCV)*, 2026.
- 2) **G. R. Matta**, T. Reddypalli, and K. Mitra, "FLAVORS: Flare-Free Novel View Synthesis with Unified Implicit and Explicit Scene Representations," *In Preparation for Journal Submission*
- 3) Sivakumar K.S., Rahman Mohammad Daniyalur, and **G. R. Matta**, "PySIFT: GPU-Resident Deterministic SIFT for Deep Learning Vision Pipelines," *arXiv preprint arXiv:2605.17869*, 2026.
- 4) M. Chandak, S. S. T. Kuppa, Rahul, **G. R. Matta**, V. Gupta, and K. Mitra, "FlareGS: 4D Flare Removal Using Gaussian Splatting for Urban Scenes," *IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2025.
- 5) S. Sharma, **G. R. Matta**, and K. Mitra, "SPC to 3D: Novel View Synthesis from Binary SPC via I2I Translation," *IEEE International Conference on Image Processing (ICIP)*, 2025.
- 6) A. Kisar, S. Shekhar, **G. R. Matta**, and K. Mitra, "A Hybrid Dataset and Depth-Guided Transformer for Nighttime Flare Removal," *16th Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)*, 2025.
- 7) **G. R. Matta**, T. Reddypalli, and K. Mitra, "BeSplat: Gaussian Splatting from a Single Blurry Image and Event Stream," *IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, 2025.
- 8) **G. R. Matta**, R. Siddartha, R. S. V. Girish, S. Sharma, and K. Mitra, "GN-FR: Generalizable Neural Radiance Fields for Flare Removal," *British Machine Vision Conference (BMVC)*, 2024.
- 9) N. Jain, **G. R. Matta**, and K. Mitra, "Towards Realistic Underwater Dataset Generation and Color Restoration," *13th Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)*, 2022.

Selected Research Projects

GN-FR
Implicit Neural Radiance Fields for Flare Rejection

- Framed view-dependent lens flare removal as an implicit multi-view reconstruction problem using NeRF architectures.
- Designed specialized mask generation and sampling pipelines to explicitly identify and filter corrupted visual evidence.
- **Tech Stack:** PyTorch, NeRF architectures, Ray-marching pipelines, Custom Sampling Masks

FLAVORS
Flare-Free Real-Time Explicit Novel View Synthesis

- Transitioned multi-view flare removal into the explicit, real-time 3D Gaussian Splatting paradigm.
- Formulated the Flare-Robust Exposure Alignment Loss (FREAL) to decouple transient aberrations from stable geometry.

- **Tech Stack:** Python, CUDA Kernels, 3D Gaussian Splatting, Custom Multi-view Loss Functions

PRISM3D
Motion-Blur Robust 3D Scene Modeling

- Developed an explicit 3DGS framework using deep dense tracking to recover global topology under extreme blur.
- Adopted a probabilistic MCMC formulation for primitive densification paired with continuous Bézier trajectories.
- **Tech Stack:** PyTorch, 3DGS, Deep Dense Tracking Modules, MCMC Optimization, Bézier Formulations

BeSplat
Single-Image Spatial-Temporal Radiance Field Recovery

- First framework recovering a sharp 3D radiance field from a single severely motion-blurred image and event stream.
- Jointly optimized explicit 3D representations and continuous camera ego-motion trajectories via Bézier formulations.
- **Tech Stack:** Python, PyTorch, Event-based Vision Data Systems, Continuous Ego-motion Trajectories

Selected Talks & Industrial Demos

- **Technical Pitch**, "Resilient Radiance Fields for Immersive XR Environments," Samsung R&D Institute India (SRI-B), IIT Madras.
  Demonstrated robust 3D scene reconstruction and explicit novel view synthesis systems directly to Dr. Lokesh Boregowda (Senior Director) and Pawan Prasad (Head of XR Interactions).
- **Interactive Demo**, "Robust 3D Reconstruction under Severe Optical and Temporal Degradations," Meta Reality Labs Partnership Review, IIT Madras.
  Presented explicit scene modeling frameworks directly to Dr. Prince Gupta (Director of Product Management, Meta).
- **Interactive Demo**, "Resilient Radiance Fields: Robust 3D Scene Reconstruction Under Severe Optical and Temporal Degradations," Dolby Laboratories Advanced Technology Group (ATG) Leadership Visit, IIT Madras.
  Presented real-time flare removal (*FLAVORS*) and blur-robust 3D reconstruction (*PRISM3D*) workflows directly to Dr. Guan-Ming Su (Director of Sight Lab).
- **Curated Speaker**, "Clearer, Sharper, Better: Pushing Radiance Fields to the Next Level," *EE Research Symposium*, IIT Madras (March 2025).
  Selected as one of 15 research scholars department-wide to present doctoral advances in Neural Radiance Fields (NeRFs) and 3D Gaussian Splatting.

Technical Skills

- **Programming:** Python, C++, CUDA, MATLAB
- **Deep Learning:** PyTorch, TensorFlow
- **Computer Vision:** 3DGS, NeRFs, SLAM, SfM, COLMAP
- **Tools:** Docker, Linux, Git, Blender, GTA-V

Mentorship & Academic Leadership

Thesis & Project Mentor — *IIT Madras*

- **Research Supervision:** Mentored 15+ students (2 M.Tech, 1 DDP, 4 B.Tech, 5 UGRC, 3 Research Interns) on computational imaging and 3D vision research projects.
- **Course Project Guidance:** Guided multiple graduate student cohorts through the successful completion of deep learning vision pipelines.

Lead Teaching Assistant & Lab Coordinator — *IIT Madras*

- **Curriculum Delivery:** Served as Lead TA for *Modern Computer Vision (EE5178)* and *Computational Photography*. Delivered lectures, designed assignments, formulated and guided the course projects.
- **Large-Scale Management:** Directed course operations and logistical infrastructure for a cohort of ∼250 students and a team of 20–30 Teaching Assistants within the *Signals and Systems* core curriculum.
- **Lab Operations:** Coordinated daily operational workflows, server configurations, and procurement logistics for high-performance NVIDIA RTX 6000 ADA GPU systems in a 30-member lab.

Professional Service

1) **Official Reviewer:** Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP).

2) **Sub-Reviewer (Assisted Peer Review):** Assisted peer review for 15+ submissions across top-tier venues including CVPR, ICCV, ECCV, NeurIPS, WACV, AAAI, ICCP, and journals including IEEE TCI, IEEE OJSP and other major venues.

Achievements/Awards

- Qualified JRF-NET (2014).
- GATE AIR 1352 (ECE, 2016).
- First Rank in class, B.Tech (ECE), RGUKT Nuzvid.
- Second Rank, M.Tech (ETCE), IIEST Shibpur.
- Core Researcher for an ICSR-funded Exploratory Research Project (6.81 Lakhs (INR)) on "Wide Field-of-view and Dense 3D Reconstruction of Underwater Archeological Sites," selected from a highly competitive pool (top 30% of 95 proposals).
- Selected for the competitive 6-year integrated Gifted Rural Youth Program, AP Government.

Industrial Training

May 2016 – July 2016

(M.Tech / Mentor: Mr. Krishna Mahesh Karanam) *Texas Instruments India, Bangalore*

- Developed a behavioral model using Verilog-A on Cadence for a 1Gbps receiver, optimizing channel equalization with CTLE to transmit 1.25Gbps data through coaxial cable and achieved 50ps jitter.

Academic References

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